

Mr. DAVIDE CAPONE

Junior Data Scientist

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Data Scientist passionate about artificial intelligence and its industrial applications. Experienced in developing and optimizing machine learning and deep learning models for data analysis, prediction, and automation, with a strong focus on practical, scalable solutions.

EXPERIENCE

Present Jul 2025	Junior Data Scientist, Danieli S.p.A., Buttrio (UD), Italy ➢ Support multiple company streams by producing reports and developing customized dashboards for analysis and trend monitoring. ➢ Designed and implemented a tailored computer vision system to support decision-making processes. Microsoft Power BIDataflowsSQL
Dec 2024 May 2025	Deep Learning Engineer (Intern), Danieli Automation S.p.A., Buttrio (UD), Italy ➢ Researched, developed, and optimized deep learning models for industrial computer vision applications. ➢ Utilized modern ML frameworks to ensure efficient and scalable implementations. ➢ Designed and implemented advanced object tracking algorithms to solve industrial challenges. Object TrackingModel Deployment
Oct 2023 Aug 2024	Computer Science Teacher, ISIS Paschini Linussio / ISIS Raimondo D'Aronco, Udine (UD), Italy ➢ Taught programming and computer science to high school students. ➢ Designed lessons combining theory with practical exercises. CommunicationTime management
Oct 2022 Dec 2022	Deep Learning Engineer (Intern), Danieli Research Center, Buttrio (UD), Italy ➢ Applied computer vision techniques, including camera calibration and homography estimation, for object detection and localization in industrial settings. ➢ Researched and implemented state-of-the-art algorithms to optimize detection and localization in steel slab storage environments. ➢ Designed, implemented, and tested convolutional neural networks (CNNs) for object detection and localization. ➢ Performed model evaluation and hyperparameter tuning to identify optimal solutions for deployment. Object DetectionCamera Calibration

EDUCATION

July 2025 Oct 2023	Master's Degree in Data Science and Scientific Computing, University of Trieste, Trieste, Italy ➢ Final Grade: 110/110 — Thesis: <i>Multi-Class Multi-Object Tracking: A Comprehensive Industrial Case Study</i>. ➢ Jointly offered by the University of Trieste, University of Udine, ICTP, and SISSA. ➢ <i>Advanced Programming and Algorithmic Design, Data Management for Big Data, High Performance Computing, Numerical Analysis, Probabilistic Machine Learning, Reinforcement Learning, Statistical Methods for Data Science, Deep Learning, Information Retrieval, Data Visualization.</i>
Dec 2022 Oct 2019	Bachelor's Degree in Internet of Things, Big Data & Web, University of Udine, Udine, Italy ➢ Final Grade: 100/110 — Thesis: <i>Applications of Computer Vision for Object Detection and Localization in a Steel Yard Plant</i>. ➢ <i>Algorithms and Data Structures, Machine Learning, Data Science, Applied Statistics, Social Computing, Mathematical Analysis, Object-Oriented Programming, Cloud Computing, Software Engineering, Databases.</i>

PROJECTS

GAPC - ASTEROID OBSERVATION CATALOG

 [GitHub Repository](#)

Django-based application to catalog asteroid observations following astronomical standards. Enables researchers to store and analyze asteroid metadata efficiently.

[Django](#) [Python](#) [Database](#) [Web Development](#)

YOHO24

 [GitHub Repository](#)

Deep learning model for sound event detection inspired by the YOLO family. YOHO24 builds on the YOHO (You-Only-Hear-Once) approach by incorporating improvements introduced in later YOLO versions to enhance model efficiency and accuracy. The model was tested on standard audio segmentation datasets, and its performance was compared with the original YOHO implementation to evaluate improvements in sound event detection.

[Deep Learning](#) [Python](#) [Sound Processing](#) [PyTorch](#)

WEIGHTED MATRIX FACTORIZATION MODEL

 [GitHub Repository](#)

Developed a movie recommender system using matrix factorization techniques. The Weighted Matrix Factorization model was implemented and tested, comparing optimization methods such as SGD and WALS to improve recommendation accuracy and system efficiency.

[Python](#) [Recommender Systems](#) [Machine Learning](#) [Data Analysis](#)

TRAFFIC LIGHTS OPTIMIZATION

 [GitHub Repository](#)

Applied Reinforcement Learning techniques, including Policy Iteration and Value Iteration, to optimize traffic light policies, reducing waiting times and queue lengths.

[Reinforcement Learning](#) [Python](#) [Optimization](#) [Simulation](#)

CERTIFICATES

INTRODUCTION TO DEEP LEARNING & NEURAL NETWORKS WITH KERAS

 IBM Certificate

[Deep Learning](#) [Neural Networks](#) [TensorFlow](#) [Keras](#) [Machine Learning](#)

MACHINE LEARNING IN PRODUCTION

 DeepLearning.AI Certificate

[MLOps](#) [Data Pipelines](#) [Feature Engineering](#) [Deployment](#) [Monitoring](#) [Data Quality](#)

MACHINE LEARNING WITH PYTHON

 IBM Certificate

[Machine Learning](#) [Python](#) [Scikit-Learn](#) [Feature Engineering](#) [Predictive Modeling](#) [Regression Analysis](#)